INCH-POUND

MIL-DTL-27426/4A
6 January 2010
SUPERSEDING
MIL-DTL-27426/4
22 September 1998

DETAIL SPECIFICATION

RINGS, RETAINING, SPIRAL
(UNIFORM CROSS SECTION)
HEAVY DUTY INTERNAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-27426, listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

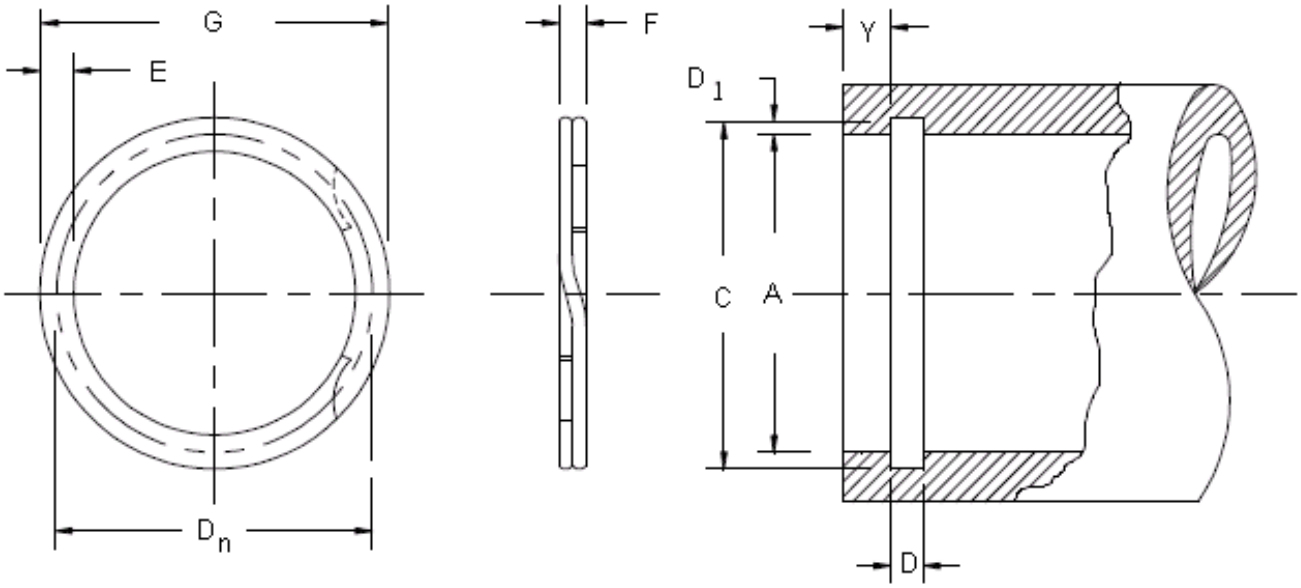


FIGURE 1 INTERNAL RING

Maximum bottom groove radius is 0.005 inches for rings up to 1.063 inches free diameter and 0.010 inches for larger rings. Ring thickness applies only to unplated rings. Add 0.002 inches to upper thickness tolerance for plated rings. Groove widths listed will accept rings without modification.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia, ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or email to dscpg&ispecomments@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:49Z
Check the source to verify that this is the current version before use.

TABLE I. HEAVY DUTY INTERNAL RETAINING RINGS.

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0100	.500	12.7	.538		.035		.045		.530		.039		.015
0101	.512	13.0	.550		.035		.045		.542		.039		.015
0102	.562	14.3	.605		.035		.055		.596	±.002	.039		.017
0103	.625	15.9	.675		.035		.055		.665		.039		.020
0104	.688	17.5	.743		.035		.065		.732		.039		.022
0105	.750	19.0	.807		.035		.065		.796		.039		.023
0106	.777	19.7	.836		.042		.075		.825		.046	+.003	.024
0107	.812	20.6	.873	+.013	.042		.075		.862		.046	-.000	.025
0108	.866	22.0	.931	-.000	.042		.075		.920		.046		.027
0109	.875	22.2	.943		.042		.085		.931	±.003	.046		.028
0110	.901	22.9	.972		.042		.085		.959		.046		.029
0111	.938	23.8	1.013		.042		.085		1.000		.046		.031
0112	1.000	25.4	1.080		.042	±.002	.085		1.066		.046		.033
0113	1.023	26.0	1.105		.042		.085	+.000	1.091		.046		.034
0114	1.062	27.0	1.138		.050		.103	-.005	1.130		.056		.034
0115	1.125	28.6	1.205		.050		.103		1.197		.056		.036
0116	1.188	30.2	1.271		.050		.103		1.262		.056		.037
0117	1.250	31.7	1.339		.050		.103		1.330		.056		.040
0118	1.312	33.3	1.406		.050		.118		1.396	±.004	.056		.042
0119	1.375	34.9	1.471		.050		.118		1.461		.056		.043
0120	1.439	36.5	1.539		.050		.118		1.528		.056		.045
0121	1.456	37.0	1.559		.050		.118		1.548		.056		.046
0122	1.500	38.1	1.605		.050		.118		1.594		.056		.047
0123	1.562	39.6	1.675		.062		.128		1.658		.068		.048
0124	1.625	41.2	1.742		.062		.128		1.725		.068		.050
0125	1.653	42.0	1.772		.062		.128		1.755		.068		.051
0126	1.688	42.8	1.810		.062		.128		1.792		.068		.052
0127	1.750	44.4	1.876	+.020	.062		.128		1.858	±.005	.068		.054
0128	1.812	46.0	1.940	-.000	.062		.128		1.922		.068		.055
0129	1.850	47.0	1.981		.062		.158		1.962		.068	+.004	.056
0130	1.875	47.6	2.008		.062		.158		1.989		.068	-.000	.057
0131	1.938	49.2	2.075		.062		.158		2.056		.068		.059
0132	2.000	50.8	2.142		.062		.158		2.122		.068		.061
0133	2.062	52.3	2.201		.078		.168		2.186		.086		.062
0134	2.125	53.9	2.267		.078		.168		2.251		.086		.063
0135	2.188	55.5	2.334		.078		.168		2.318		.086		.065
0136	2.250	57.1	2.399	+.025	.078		.168		2.382		.086		.066
0137	2.312	58.7	2.467	-.000	.078		.200		2.450		.086		.069
0138	2.375	60.3	2.535		.078		.200		2.517		.086		.071
0139	2.440	62.0	2.602		.078		.200		2.584		.086		.072
0140	2.500	63.5	2.667		.078		.200		2.648		.086	+.005	.074
0141	2.531	64.2	2.700		.078		.200	+.004	2.681	±.006	.086	-.000	.075
0142	2.562	65.1	2.733		.093		.225	-.006	2.714		.103		.076
0143	2.625	66.6	2.801		.093		.225		2.781		.103		.078
0144	2.688	68.2	2.868		.093		.225		2.848		.103		.080
0145	2.750	69.8	2.934	+.030	.093		.225		2.914		.103		.082
0146	2.813	71.4	3.001	-.000	.093	±.003	.225		2.980		.103		.084
0147	2.834	71.9	3.027		.093		.225		3.006		.103		.086
0148	2.875	73.0	3.072		.093		.225		3.051		.103		.088
0149	3.000	76.1	3.204		.093		.225		3.182		.103		.091
0150	3.062	77.7	3.271		.111		.281		3.248		.120		.093
0151	3.125	79.3	3.338	+.035	.111		.281		3.315		.120		.095
0152	3.157	80.1	3.371	-.000	.111		.281		3.348		.120		.096
0153	3.250	82.5	3.470		.111		.281		3.446		.120		.098
0154	3.346	84.9	3.571		.111		.281		3.546		.120		.100

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:49Z
Check the source to verify that this is the current version before use.

TABLE I. HEAVY DUTY INTERNAL RETAINING RINGS. (CONT'D)

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0155	3.469	88.0	3.701		.111		.281	+.004	3.675		.120		.105
0156	3.500	88.8	3.736		.111		.281	-.006	3.710		.120		.105
0157	3.543	89.9	3.781		.111		.281		3.755		.120		.106
0158	3.562	90.5	3.802		.111		.281		3.776		.120		.107
0159	3.625	92.0	3.868		.111		.281		3.841		.120		.108
0160	3.750	95.2	4.002		.111		.312		3.974	±.006	.120	+.005	.112
0161	3.875	98.3	4.136	+.035	.111		.312		4.107		.120	-.000	.116
0162	3.938	99.9	4.203	-.000	.111		.312		4.174		.120		.118
0163	4.000	101.5	4.270		.111		.312		4.240		.120		.120
0164	4.125	104.8	4.369		.111	±.003	.312		4.339		.120		.120
0165	4.250	107.9	4.501		.111		.312		4.470		.120		.120
0166	4.330	109.9	4.588		.111		.312		4.556		.120		.120
0167	4.500	114.2	4.768		.111		.312		4.735		.120		.120
0168	4.625	117.4	4.899		.111		.312		4.865		.120		.120
0169	4.750	120.6	5.030		.111		.312		4.995		.120		.123
0170	5.000	126.9	5.297		.111		.312		5.260		.120		.130
0171	5.250	133.2	5.559		.127		.350		5.520		.139		.135
0172	5.375	136.5	5.690	+.050	.127		.350		5.650		.139	+.006	.135
0173	5.500	139.7	5.810	-.000	.127	±.004	.350		5.770	±.007	.139	-.000	.135
0174	5.750	145.9	6.062		.127		.350	+.004	6.020		.139		.135
0175	6.000	152.3	6.314		.127		.350	-.008	6.270		.139		.135
0176	6.250	158.8	6.576		.156		.380		6.530		.174		.140
0177	6.500	165.1	6.838		.156		.380		6.790		.174		.145
0178	6.625	168.3	6.974	+.055	.156		.380		6.925		.174		.150
0179	6.750	171.4	7.105	-.000	.156		.380		7.055		.174		.153
0180	7.000	177.8	7.366		.156		.380		7.315		.174		.158
0181	7.250	184.0	7.628		.187		.418		7.575		.209		.163
0182	7.500	190.5	7.895		.187		.418		7.840		.209		.170
0183	7.750	197.0	8.157		.187		.418		8.100		.209		.175
0184	8.000	203.3	8.419	+.070	.187		.418		8.360	±.008	.209		.180
0185	8.250	209.8	8.680	-.000	.187		.437		8.620		.209		.185
0186	8.500	216.0	8.942		.187		.437		8.880		.209		.190
0187	8.750	222.2	9.209		.187		.437		9.145		.209		.198
0188	9.000	228.8	9.471		.187	±.005	.437		9.405		.209		.203
0189	9.250	235.0	9.737		.187		.437		9.669		.209	+.008	.210
0190	9.500	241.5	10.000		.187		.500		9.930		.209	-.000	.215
0191	9.750	248.0	10.260		.187		.500		10.189		.209		.220
0192	10.000	254.0	10.523		.187		.500		10.450		.209		.225
0193	10.250	260.8	10.786		.187		.500		10.711		.209		.230
0194	10.500	266.7	11.047		.187		.500		10.970		.209		.235
0195	10.750	273.1	11.313		.187		.500		11.234		.209		.242
0196	11.000	279.4	11.575		.187		.500		11.495		.209		.247
0197	11.250	285.8	11.838		.187		.500		11.756		.209		.253
0198	11.500	292.1	12.102	+.120	.187		.562	+.005	12.018	±.010	.209		.259
0199	11.750	298.5	12.365	-.000	.187		.562	-.010	12.279		.209		.265
0200	12.000	304.8	13.628		.187		.562		12.540		.209		.270
0201	12.250	311.2	12.891		.187		.562		12.801		.209		.276
0202	12.500	317.5	13.154		.187		.562		13.063		.209		.282
0203	12.750	323.9	13.417		.187		.562		13.324		.209		.287
0204	13.000	330.2	13.680		.187		.662		13.585		.209		.293
0205	13.250	336.6	13.943		.187		.662		13.846	±.012	.209		.298
0206	13.500	342.9	14.207		.187		.662		14.108		.209		.304
0207	13.750	349.3	14.470	+.140	.187		.662		14.369		.209		.310
0208	14.000	355.6	14.732	-.000	.187		.662		14.630		.209		.315
0209	14.250	362.0	14.995		.187		.662		14.891		.209		.321
0210	14.500	368.3	15.259		.187		.750	+.005	15.153	±.012	.209	+.008	.327

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:49Z
Check the source to verify that this is the current version before use.

TABLE I. HEAVY DUTY INTERNAL RETAINING RINGS. (CONT'D)

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G		THICKNESS F		RADIAL WALL E		DIAMETER C		WIDTH D		NOMINAL GROOVE DEPTH D_1
				TOL		TOL		TOL		TOL		TOL	
0211	14.750	374.7	15.522	+.140	.187	±.005	.750	-.010	15.414		.209	-.000	.332
0212	15.000	381.0	15.785	-.000	.187		.750		15.675		.209		.338

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AR
Navy - OS
Air Force - 99

Preparing activity:
DLA-IS

(Project 5325-2010-001)

Review activities:
Army – AV, MI
Air Force – 11, 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:49Z
Check the source to verify that this is the current version before use.